Interaktioner og modelevaluering

Sören Möller

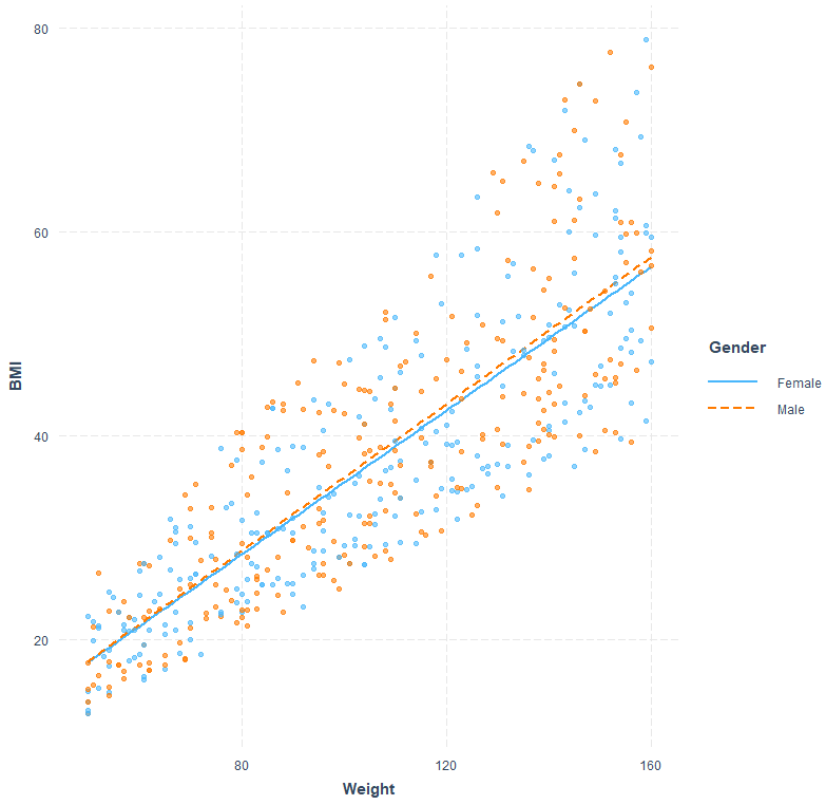
Institut for Sundhedstjenesteforskning
Syddansk Universitet

19. februar 2026

Interaktioner

- Ofte oplever vi at an sammenhæng vi måler er forskellig mellem to grupper
  - Behandling virker bedre i unge end i gamle
  - Sammenhæng mellem biomarkører og risiko højere i kvinder end i mænd
- Hvis denne forskel er kausalt betegner vi den som *effektmodifikation*
- Statistisk betegner vi den som *interaktion*
  - Den modelleres ved at inkludere et interaktionsled i regressionsmodeller

Figur: Ingen interaktion




```
library(interactions)
linreg <- lm(BMI ~ Weight*factor(Gender), data=bmi)
interact_plot(linreg, pred = Weight, modx = Gender, plot.points = TRUE)
```

Lineær regression i R: Ingen interaktion

```
> linreg <- lm(BMI ~ Weight*factor(Gender), data=bmi)
> summary(linreg)
Call:
lm(formula = BMI ~ Weight * factor(Gender), data = bmi)
Residuals:
    Min      1Q  Median      3Q     Max
-16.672  -5.923  -1.263   5.075  22.928
Coefficients:
                           Estimate Std. Error t value Pr(>|t|)
(Intercept)                0.076023   1.658228   0.046    0.963
Weight                     0.353063   0.014980  23.570   <2e-16 ***
factor(Gender)Male        -0.253505   2.415213  -0.105    0.916
Weight:factor(Gender)Male  0.007463   0.021791   0.342    0.732
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
Residual standard error: 7.869 on 496 degrees of freedom
Multiple R-squared:  0.6844,        Adjusted R-squared:  0.6825
F-statistic: 358.6 on 3 and 496 DF,  p-value: < 2.2e-16
> confint(linreg)
                               2.5 %     97.5 %
(Intercept)               -3.18199449 3.33404012
Weight                     0.32363187 0.38249440
factor(Gender)Male        -4.99881542 4.49180603
Weight:factor(Gender)Male -0.03535052 0.05027683
```

Figur: Interaktion

```
library(interactions)
linreg <- lm(BMI2 ~ Weight*factor(Gender), data=bmi)
interact_plot(linreg, pred = Weight, modx = Gender, plot.points = TRUE)
```

Lineær regression i R: Interaktion

```
> linreg <- lm(BMI2 ~ Weight*factor(Gender), data=bmi)
> summary(linreg)
Call:
lm(formula = BMI2 ~ Weight * factor(Gender), data = bmi)
Residuals:
    Min      1Q  Median      3Q     Max
-16.672  -5.923  -1.263   5.075  22.928
Coefficients:
                          Estimate Std. Error t value Pr(>|t|)
(Intercept)                0.07602    1.65823   0.046    0.963
Weight                     0.35306    0.01498  23.570  < 2e-16 ***
factor(Gender)Male        -0.25350    2.41521  -0.105    0.916
Weight:factor(Gender)Male  0.10746    0.02179   4.932 1.11e-06 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
Residual standard error: 7.869 on 496 degrees of freedom
Multiple R-squared:  0.7704,        Adjusted R-squared:  0.769
F-statistic: 554.7 on 3 and 496 DF,  p-value: < 2.2e-16
> confint(linreg)
                              2.5 %    97.5 %
(Intercept)               -3.18199449 3.3340401
Weight                     0.32363187 0.3824944
factor(Gender)Male        -4.99881542 4.4918060
Weight:factor(Gender)Male  0.06464948 0.1502768
```

Lineær regression i R: Effekt i hver gruppe

```
> linreg <- lm(BMI2 ~ Weight:factor(Gender)+factor(Gender), data=bmi)
> summary(linreg)
Call:
lm(formula = BMI2 ~ Weight:factor(Gender) + factor(Gender), data = bmi)
Residuals:
    Min      1Q  Median      3Q     Max
-16.672  -5.923  -1.263   5.075  22.928
Coefficients:
                            Estimate Std. Error t value Pr(>|t|)
(Intercept)                  0.07602    1.65823   0.046    0.963
factor(Gender)Male          -0.25350    2.41521  -0.105    0.916
Weight:factor(Gender)Female  0.35306    0.01498  23.570   <2e-16 ***
Weight:factor(Gender)Male    0.46053    0.01583  29.100   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
Residual standard error: 7.869 on 496 degrees of freedom
Multiple R-squared:  0.7704,        Adjusted R-squared:  0.769
F-statistic: 554.7 on 3 and 496 DF,  p-value: < 2.2e-16
> confint(linreg)
                                 2.5 %    97.5 %
(Intercept)                 -3.1819945 3.3340401
factor(Gender)Male          -4.9988154 4.4918060
Weight:factor(Gender)Female  0.3236319 0.3824944
Weight:factor(Gender)Male    0.4294327 0.4916199
```

Logistisk regression i R: Uden interaktion

```
> load(file="forkoelelse.RData")
> logreg <- glm(factor(forkoelet_foraeldre) ~ factor(institution) + factor(samboende)
+ alder, data = forkoelelse, family = "binomial")
> summary(logreg)
Call:
glm(formula = factor(forkoelet_foraeldre) ~ factor(institution) +
    factor(samboende) + alder, family = "binomial", data = forkoelelse)
Coefficients:
                     Estimate Std. Error z value Pr(>|z|)
(Intercept)          -5.16141    1.05039  -4.914 8.93e-07 ***
factor(institution)1  0.93314    0.19358   4.820 1.43e-06 ***
factor(samboende)1    0.01263    0.16719   0.076     0.94
alder                 0.18074    0.03277   5.515 3.48e-08 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
(Dispersion parameter for binomial family taken to be 1)
    Null deviance: 1268.4  on 999  degrees of freedom
Residual deviance: 1234.4  on 996  degrees of freedom
AIC: 1242.4
Number of Fisher Scoring iterations: 4
> exp(coef(logreg))
         (Intercept) factor(institution)1   factor(samboende)1                alder
         0.005733582          2.542486117          1.012714086          1.198104591
> exp(confint(logreg))
Waiting for profiling to be done...
                           2.5 %     97.5 %
(Intercept)          0.0007142304 0.04402554
factor(institution)1 1.7448163565 3.72874707
factor(samboende)1   0.7271001031 1.40142413
alder                1.1243813655 1.27865254
```

Logistisk regression i R: Interaktion

```
> logreg <- glm(factor(forkoelet_foraeldre) ~ factor(institution)*factor(samboende)
+ alder, data = forkoelelse, family = "binomial")
> summary(logreg)
Call:
glm(formula = factor(forkoelet_foraeldre) ~ factor(institution) *
    factor(samboende) + alder, family = "binomial", data = forkoelelse)

Coefficients:
                                        Estimate Std. Error z value Pr(>|z|)
(Intercept)                             -4.85215    1.05929  -4.581 4.64e-06 ***
factor(institution)1                     0.33441    0.32727   1.022   0.3069
factor(samboende)1                      -0.37985    0.24565  -1.546   0.1220
alder                                    0.18080    0.03278   5.515 3.48e-08 ***
factor(institution)1:factor(samboende)1  0.76085    0.33644   2.261   0.0237 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
(Dispersion parameter for binomial family taken to be 1)
    Null deviance: 1268.4  on 999  degrees of freedom
Residual deviance: 1229.2  on 995  degrees of freedom
AIC: 1239.2
Number of Fisher Scoring iterations: 4
> exp(coef(logreg))
(Intercept) factor(institution)1 factor(samboende)1 alder
0.00781156 1.39711479            0.68396453         1.19817995
factor(institution)1:factor(samboende)1
2.14010136
> exp(confint(logreg))
Waiting for profiling to be done...
                                          2.5 %     97.5 %
(Intercept)                             0.0009576872 0.06114132
factor(institution)1                    0.7341475042 2.65474704
factor(samboende)1                      0.4179613694 1.09773607
alder                                   1.1244065045 1.27874805
factor(institution)1:factor(samboende)1 1.1094720764 4.15653850
```

Logistisk regression i R: Effekt i hver gruppe

```
> logreg <- glm(factor(forkoelet_foraeldre) ~ factor(institution):factor(samboende)
+ factor(samboende) + alder, data = forkoelelse, family = "binomial")
> summary(logreg)
Call:
glm(formula = factor(forkoelet_foraeldre) ~ factor(institution):factor(samboende) +
    factor(samboende) + alder, family = "binomial", data = forkoelelse)
Coefficients:
                                        Estimate Std. Error z value Pr(>|z|)
(Intercept)                             -4.85215    1.05929  -4.581 4.64e-06 ***
factor(samboende)1                      -0.37985    0.24565  -1.546    0.122
alder                                    0.18080    0.03278   5.515 3.48e-08 ***
factor(institution)1:factor(samboende)0  0.33441    0.32727   1.022    0.307
factor(institution)1:factor(samboende)1  1.09526    0.20720   5.286 1.25e-07 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
(Dispersion parameter for binomial family taken to be 1)
    Null deviance: 1268.4  on 999  degrees of freedom
Residual deviance: 1229.2  on 995  degrees of freedom
AIC: 1239.2
Number of Fisher Scoring iterations: 4
> exp(coef(logreg))
(Intercept) factor(samboende)1 alder      factor(institution)1:factor(samboende)0
0.00781156  0.68396453         1.19817995 1.39711479
factor(institution)1:factor(samboende)1
2.98996726
> exp(confint(logreg))
Waiting for profiling to be done...
                                            2.5 %     97.5 %
(Intercept)                             0.0009576872 0.06114132
factor(samboende)1                      0.4179613694 1.09773607
alder                                   1.1244065045 1.27874805
factor(institution)1:factor(samboende)0 0.7341475042 2.65474704
factor(institution)1:factor(samboende)1 1.9985529752 4.50543486
```

Additive of multiplikative interaktioner

- Der findes to typer interaktioner for logistisk, log-binomial, Poisson-, Cox-regression:
  - Additiv: Risikoen er højere/lavere hvis vi kigger på den på risiko-skala
  - Eksempel: Din risiko er 20% højere af at ryge, 5% højere af at være en mand, men 30% højere af at være en rygende mand
  - Multiplikativ: OR/RR/... er højere lavere hvis man har begge faktorer
  - Eksempel: OR for rygnign er 1.5, OR for mænd er 1.2, men OR for mænd der ryger sammenligned med kvinder der ikke ryger er 2.0
- For lineær regression findes denne forskel ikke (additiv er eneste mulighed)
- Traditionelt er det den multiplikative interaktion, der fokuseres mest på
- I øvelserne snakker i om en artikel, der kommer nærmere ind på forskellen
  - Og hvorfor det kan være godt at tage højde for begge slags interaktion

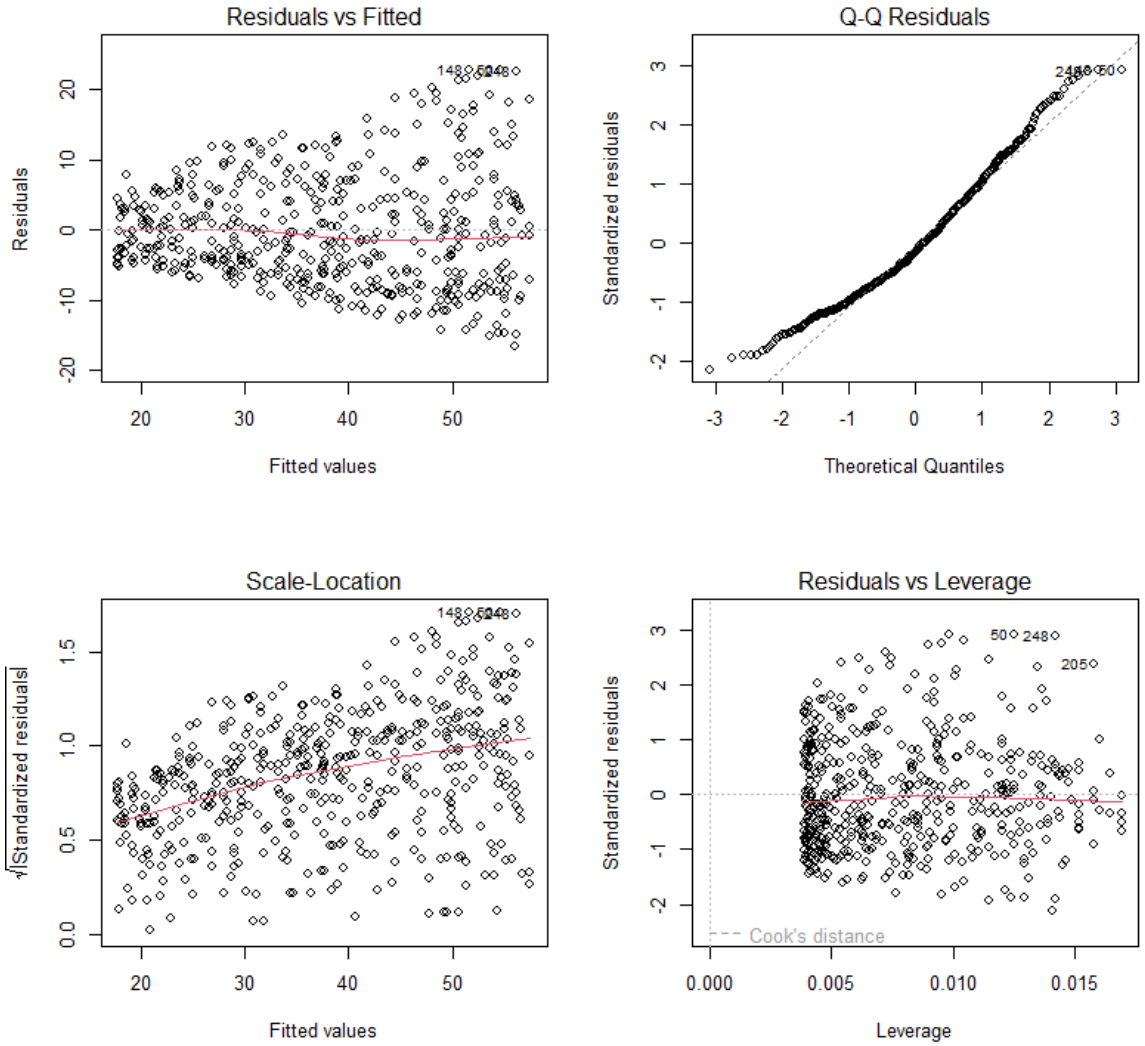
Modelantagelser

- Alle statistiske modeller har antagelser
  - Disse antagelser bør vi forholde os til
  - Og tjekke de antagelser, der kan tjekkes
- De fleste modeller antager uafhængige observationer
  - Dette kan ikke tjekkes på data, men må overvejes baseret på studiedesign
  - Der findes metoder til at håndtere ikke-uafhængige data
    - Mixed-effects modeller, generaliserede estimationsligninger, ...
    - Disse kommer vi ikke ind på i dette kursus

Modelantagelser: Lineær regression

- Uafhængige observationer
- Normalfordelte residualer
  - Tjekkes med kvantil-kvantil-plot (kan også bruge Shapiro-Wilks test)
- Linearitet af sammenhænge
  - Hvis dette ikke er opfyld, kan det skyldes en manglende interaktion
  - Eller for numeriske forklarende variable: En ikke-lineær sammenhæng
- Varianshomogenitet
  - Variationen bør ikke afhænge af størrelsen på data
- Ikke nogle ekstreme målinger, der dominerer resultatet

Figur: Modelantagelser: Lineær regression




```
linreg <- lm(BMI ~ Weight*factor(Gender), data=bmi)
par(mfrow = c(2, 2))
plot(linreg)
```

Modelantagelser: Logistisk regression

- Uafhængige observationer
- (Log)linearitet af sammenhænge
  - Hvis dette ikke er opfyld, kan det skyldes en manglende interaktion
  - Eller for numeriske forklarende variable: En ikke-lineær sammenhæng
  - Svært at tjekke grafisk, men muligt at tjekke for interaktioner
- Det primære vi tjekker et om modellens forudsigelser svarer overens med data
  - *goodnes-of-fit*-tests
  - Typisk: Hosmer-Lemeshow
  - Foretrukken: Omnibus (mere robust)
  - Ikke-signifikant betyder at modellen opfører sig pænt

Modeltjek logistisk regression i R

```
> library("performance")
> performance_hosmer(logreg, n_bins = 10)
# Hosmer-Lemeshow Goodness-of-Fit Test

  Chi-squared: 3.195
           df: 8
      p-value: 0.922

Summary: model seems to fit well.
>
> library(rms)
> logreg.res <- lrm(factor(Outcome) ~ Glucose, data =  diabetes, y = TRUE, x = TRUE)
> residuals(logreg.res, type = "gof")
Sum of squared errors     Expected value|H0                    SD                     Z
         132.57002132          134.31487580            0.60734464           -2.87292314
          P
        0.00406693
```

Hosmer-Lemenshow er den klassisk mest udbredte, men meget følsom over for antal kasser (bins). Derfor anbefales den nederste ("omnibustesten").

Modelantagelser: Poissonregression

- Udfaldet er et antal
- Observationerne er uafhængige
- Spredningen er lig med middelværdien (betinget på kovariate)
  - Det er typisk denne antagelse, der giver problemer
  - Overdispersion: Varians større end spredning (betinget på forklarende variable)
  - Hvis dette er et problem, bruger vi negativ binomial regression i stedet for Poissonregression

Modelantagelser: Poissonregression i R

```
> modeltr <- glm(Incidence ~ State, family = poisson, offset = log(Population),
data = breastcancer)
> library(AER)
> dispersiontest(modeltr)
        Overdispersion test
data:  modeltr
z = -2.4764e+15, p-value = 1
alternative hypothesis: true dispersion is greater than 1
sample estimates:
   dispersion
-2.220446e-16
> library(MASS)
> modeltr2 <- glm.nb(Incidence ~ State+ offset(log(breastcancer$Population)),
data = breastcancer,init.theta=1)
> summary(modeltr2)
Call:
glm.nb(formula = Incidence ~ State + offset(log(breastcancer$Population)),
    data = breastcancer, init.theta = 5.884502677e+28, link = log)
Coefficients:
                     Estimate Std. Error   z value Pr(>|z|)
(Intercept)        -6.6214661  0.0027902 -2373.147  < 2e-16 ***
StateAlabama        0.0961544  0.0079151    12.148  < 2e-16 ***
StateAlaska        -0.1047605  0.0219994    -4.762 1.92e-06 ***
...
(Dispersion parameter for Negative Binomial(5.884503e+28) family taken to be 1)
    Null deviance:  1.3877e+06  on 49  degrees of freedom
Residual deviance: -2.2070e-09  on  0  degrees of freedom
AIC: 102
Number of Fisher Scoring iterations: 20
              Theta:  5.884503e+28
          Std. Err.:  2.671374e+21
 2 x log-likelihood:  0
```

Overblik over metoder

Exposure→Outcome/response

Exposure(x)	Response(y)				
	binary	r categories	counts	continuous	survival time
binary (yes/no)	2 × 2 table χ^2-test Odds-ratio/rel. risk	r × 2 table χ^2_{r-1}-test Odds-ratio/rel. risk	2 rates diff in rates	comparison of 2 means *t*-test, Wilcoxon diff in means	Kaplan-Meier curves log-rank test Hazard ratio
c groups	2 × c table χ^2_{c-1}-test Odds-ratio/rel. risk	r × c table $\chi^2_{(r-1)\times(c-1)}$-test Odds-ratio/rel. risk	poisson Incidence rate ratio	comparison of c means multiple regression means	Cox-regression Hazard ratio
continuous and adjustment for covariates	logistic regression	ordinal regression proportional odds multinomial regression	poisson regression negative binomial regression	regression multiple lin regression	Cox-regression competing risk regression

- For grouped observations, eg. repeated measurements, models with random effects extends above models: Generalized linear mixed models

Spørgsmål?